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United States Patent	12416957
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Teeter; Victor et al.

PoE power aggregation system

Abstract

A PoE power aggregation system includes a PoE power aggregation device coupling a powering device to a powered device. The PoE power aggregation device includes first and second powering device connectors coupled to first and second powering device ports on the powering device, respectively, and a powered device connector coupled to a powered device port on the powered device. The PoE power aggregation device receives data from the powering device via the first powering device connector, and transmits the data to the powered device through the powered device connector. The PoE power aggregation device also receives first and second power from the powering device via the first powering device connector and the second powering device connector, respectively, and transmits the first and second power to the powered device through the powered device connector along with the data.

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Appl. No.: **18/240082**

Filed: **August 30, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20250076947 A1	Mar. 06, 2025

Publication Classification

Int. Cl.: **G06F1/26** (20060101); **H02J1/10** (20060101); **H04L12/10** (20060101); **H04L25/02** (20060101)

U.S. Cl.:

CPC **G06F1/266** (20130101); **H02J1/10** (20130101); **H04L12/10** (20130101); **H04L25/02** (20130101);

Field of Classification Search

CPC: G06F (1/263); G06F (1/266); H04L (12/10); H04L (12/40045); H04L (25/02); H02J (1/10)

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Primary Examiner: Cleary; Thomas J.

Background/Summary

BACKGROUND

(1) The present disclosure relates generally to information handling systems, and more particularly to powering a Power over Ethernet (PoE) powered device by aggregating power received from multiple ports on a powering information handling system.

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users

to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

(3) Information handling systems such as switch devices and/or other powering computing devices known in the art sometimes utilize Power over Ethernet (PoE) techniques to power powered devices connected to their ports via Ethernet cables. As will be appreciated by one of skill in the art in possession of the present disclosure, such PoE techniques allow electrical power to be transmitted by a powering computing device along with data signals over the same Ethernet cable to a powered device, and in specific examples may enable Internet of Things (IoT) devices and other network devices such as Internet Protocol (IP) cameras, wireless access points, and Voice over IP (VoIP) phones to receive power directly through a single Ethernet cable through which it sends and receives data. As such, the installation and management of such network devices is simplified by eliminating the need for separate power cabling and allowing power distribution to be centralized through the switch device, thus reducing cabling “clutter” and enabling relatively easier IoT/network device deployment, particularly in situations in which power outlets or other power sources are not available or abundant.

(4) However, the power requirements of PoE powered devices continues to increase. For example, advanced PoE standards such as the PoE+ standard described in Institute of Electrical and Electronics Engineers (IEEE) 802.3at and the Universal PoE (UPoE) standard described in IEEE 802.3bt have been developed to deliver levels of electrical power that are higher than those available using conventional PoE standards like those described in IEEE 802.3af, and that are designed to support a wider range of powered devices that require additional power for enhanced performance, features, and/or functionality by providing 30 watts, 60 watts, or more power via an Ethernet cable (e.g., as opposed to the 15.4 watts of power available via the conventional PoE standards described above). As such, switch devices and/or other powering devices operating according to conventional PoE standards may be unable to power (or properly power) powered devices operating according to the PoE+ or UPoE standards discussed above, resulting in increased costs for users that wish to implement such powered devices in their networks and thus must purchase new switch/powering devices, or inhibiting the adoption of such powered devices by such users.

(5) Accordingly, it would be desirable to provide a PoE power provisioning system that addresses the issues discussed above.

SUMMARY

(6) According to one embodiment, a Power over Ethernet (PoE) power aggregation device includes a chassis; a first powering device connector that is included on the chassis; a second powering device connector that is included on the chassis; a powered device connector that is included on the chassis; a data transmission subsystem that is included in the chassis and coupled to the first powering device connector and the first powered device connector, wherein the data transmission subsystem is configured to: receive data from a powering device through the first powering device connector; and transmit the data to a powered device through the powered device connector; and a power transmission subsystem that is included in the chassis and coupled to the first powering device connector, the second powering device connector, and the first powered device connector, wherein the power transmission subsystem is configured to: receive first power from the powering

device through the first powering device connector; receive second power from the powering device through the second powering device connector; and transmit the first power and the second power to the powered device through the powered device connector along with the data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic view illustrating an embodiment of an Information Handling System (IHS).

(2) FIG. **2** is a schematic view illustrating an embodiment of powering devices and a powered device that may be included in a networked system that may provide the PoE power aggregation system of the present disclosure.

(3) FIG. **3** is a schematic view illustrating an embodiment of a PoE power aggregation device that may be included in a networked system that may provide the PoE power aggregation system of the present disclosure.

(4) FIG. **4**A is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **3**.

(5) FIG. **4**B is a schematic front view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A.

(6) FIG. **4**C is a schematic front view illustrating another embodiment of the PoE power aggregation device of FIG. **4**A.

(7) FIG. **5**A is a schematic top view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A with a first powering device connector configuration.

(8) FIG. **5**B is a schematic front view illustrating an embodiment of the PoE power aggregation device of FIG. **5**A with the first powering device connector configuration.

(9) FIG. **5**C is a schematic front view illustrating an embodiment of the PoE power aggregation device of FIG. **5**A having its connector configuration changed from the first powering device connector configuration to a second powering device connector configuration.

(10) FIG. **5**D is a schematic front view illustrating an embodiment of the PoE power aggregation device of FIG. **5**A with the second powering device connector configuration.

(11) FIG. **6**A is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A with reconfigurable powering device connectors provided in a first powering device connector configuration.

(12) FIG. **6**B is a schematic front view illustrating an embodiment of the PoE power aggregation device of FIG. **6**A with reconfigurable powering device connectors provided in a second powering device connector configuration.

(13) FIG. **6**C is a schematic front view illustrating an embodiment of the first powering device connector configuration of the PoE power aggregation device of FIG. **6**A.

(14) FIG. **6**D is a schematic front view illustrating an embodiment of the second powering device connector configuration of the PoE power aggregation device of FIG. **6**B.

(15) FIG. **6**E is a schematic front view illustrating an embodiment of the first powering device connector configuration of the PoE power aggregation device of FIG. **6**A.

(16) FIG. **6**F is a schematic front view illustrating an embodiment of the second powering device connector configuration of the PoE power aggregation device of FIG. **6**B.

(17) FIG. **7**A is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A with reconfigurable powering device connectors provided in a first powering device connector configuration.

(18) FIG. **7**B is a schematic front view illustrating an embodiment of the PoE power aggregation device of FIG. **7**A with reconfigurable powering device connectors provided in a second powering

device connector configuration.
(19) FIG. **8** is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A with cabled powering device connectors.
(20) FIG. **9** is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A.
(21) FIG. **10** is a schematic view illustrating an embodiment of a rackable PoE power aggregation system including a plurality of the PoE power aggregation devices of FIG. **9**.
(22) FIG. **11** is a flow chart illustrating an embodiment of a method for aggregating PoE power from a powering device for use by a powered device.
(23) FIG. **12** is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A coupled a powering device and the powered device in the networked system of FIG. **2** to provide the PoE power aggregation system of the present disclosure.
(24) FIG. **13**A is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A that may be used during the method of FIG. **11**.
(25) FIG. **13**B is a schematic view illustrating the PoE power aggregation device of FIG. **13**A operating during the method of FIG. **11**.
(26) FIG. **14**A is a schematic view illustrating an embodiment of the PoE power aggregation device of FIG. **4**A operating during the method of FIG. **11**.
(27) FIG. **14**B is a schematic view illustrating the PoE power aggregation device of FIG. **14**A operating during the method of FIG. **11**.
(28) FIG. **15** is a schematic view illustrating the PoE power aggregation device of FIG. **14**A operating during the method of FIG. **11**.

DETAILED DESCRIPTION

(29) For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.
(30) In one embodiment, IHS **100**, FIG. **1**, includes a processor **102**, which is connected to a bus **104**. Bus **104** serves as a connection between processor **102** and other components of IHS **100**. An input device **106** is coupled to processor **102** to provide input to processor **102**. Examples of input devices may include keyboards, touchscreens, pointing devices such as mouses, trackballs, and trackpads, and/or a variety of other input devices known in the art. Programs and data are stored on a mass storage device **108**, which is coupled to processor **102**. Examples of mass storage devices may include hard discs, optical disks, magneto-optical discs, solid-state storage devices, and/or a variety of other mass storage devices known in the art. IHS **100** further includes a display **110**, which is coupled to processor **102** by a video controller **112**. A system memory **114** is coupled to processor **102** to provide the processor with fast storage to facilitate execution of computer programs by processor **102**. Examples of system memory may include random access memory

(RAM) devices such as dynamic RAM (DRAM), synchronous DRAM (SDRAM), solid state memory devices, and/or a variety of other memory devices known in the art. In an embodiment, a chassis **116** houses some or all of the components of IHS **100**. It should be understood that other buses and intermediate circuits can be deployed between the components described above and processor **102** to facilitate interconnection between the components and the processor **102**.
(31) Referring now to FIG. **2**, an embodiment of a powering devices and a powered device that may be included in a networked system that may provide the PoE power aggregation system of the present disclosure are illustrated. In the illustrated embodiment, a plurality of powering devices **200**, **202**, and up to **204** may be included in the networked system that provides the PoE power aggregation system of the present disclosure, and any or each of the powering devices **200-204** may be provided by the IHS **100** discussed above with reference to FIG. **1**, and/or may include some or all of the components of the IHS **100**. One of skill in the art in possession of the present disclosure will recognize that the specific examples provided below describe the powering devices **202-206** as being provided by switch devices, but will recognize that powering devices provided in the PoE power aggregation system of the present disclosure may include any powering devices (e.g., server devices, storage systems, and/or other computing devices that would be apparent to one of skill in the art in possession of the present disclosure) that may be configured to operate similarly as the powering devices **200-204** discussed below.
(32) In the illustrated embodiment, each of the powering devices **200**, **202**, and up to **204** includes a plurality of powering device ports **200***a*-**200***b*, **202***a*-**202***b*, and up to **204***a*-**204***b*, respectively, and while only two powering device ports are illustrated as being included on each of the powering devices **200-204** for the purposes of the specific examples described below, one of skill in the art in possession of the present disclosure in the art in possession of the present disclosure will appreciate how the powering devices may include many more powering device ports while remaining within the scope of the present disclosure as well. As discussed below, the powering device ports may be provided by Ethernet ports (e.g., female Ethernet ports), although other powering devices ports are envisioned as falling within the scope of the present disclosure as well.
(33) In the illustrated embodiment, a powered device **206** may also be included in the networked system that provides the PoE power aggregation system of the present disclosure, and may be provided by the IHS **100** discussed above with reference to FIG. **1**, and/or may include some or all of the components of the IHS **100**. One of skill in the art in possession of the present disclosure will recognize that the specific examples provided below describe the powered device **206** as being provided by IP cameras, wireless access points, and VoIP phones, and/or other IoT/network devices known in the art, but will recognize that powered devices provided in the PoE power aggregation system of the present disclosure may include any powered devices that may be configured to operate similarly as the powered device **206** discussed below. In the illustrated embodiment, the powered device **206** includes a powered device port **206***a*. As discussed below, the powered device port may be provided by an Ethernet port (e.g., female Ethernet port), although other powered devices ports are envisioned as falling within the scope of the present disclosure as well. However, while a specific example of components in a networked system that provides the PoE power aggregation system of the present disclosure have been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that the PoE power aggregation system of the present disclosure may include a variety of components and component configurations while remaining within the scope of the present disclosure as well.
(34) Referring now to FIG. **3**, an embodiment of a PoE power aggregation device **300** is illustrated that may couple any of the powering devices **200***a*-**200***c* to the powered device **202** discussed above with reference to FIG. **2**. In the illustrated embodiment, the PoE power aggregation device **300** includes a chassis **302** that houses the components of the PoE power aggregation device **300**, only some of which are illustrated and described below. For example, the chassis **302** may include a plurality of powering device connectors **304***a*, **304***b*, and up to **304***c*. To provide some specific

examples, two powering device connectors may be provided on the PoE power aggregation device **300** to configure the PoE power aggregation device **300** to allow a conventional PoE powering device to power some PoE+ powered devices (or configure the PoE power aggregation device **300** to allow a PoE+ powering device to power some UPoE powered devices), and three powering device connectors may be provided on the PoE power aggregation device **300** to configure the PoE power aggregation device **300** to allow a conventional PoE powering device to power some UPoE powered devices.

(35) However, while specific numbers of powering device connectors have been described, one of skill in the art in possession of the present disclosure will appreciate how different numbers of powering device connectors may be provided on the PoE power aggregation device **300** to allow a variety of powering devices to power powered devices that require more power than is available by any one port on those powering devices while remaining within the scope of the present disclosure as well. In many of the embodiments discussed below, the powering device connectors **304***a*-**304***c* are provided by Ethernet connectors, although one of skill in the art in possession of the present disclosure will appreciate how other types of powering device connectors may enable the teachings of the present disclosure and thus are envisioned as falling within its scope as well.

(36) Furthermore, the chassis **302** may also include a powered device connector **306** that is coupled to the powering device connector **304***a*, and one of skill in the art in possession of the present disclosure will appreciate how the coupling between the powered device connector **306** and the powering device connector **304***a* may be provided by cabling, traces in a circuit board that supports the powered device connector **306** and the powering device connector **304***a*, and/or other data communication subsystems that may operate as a data “passthrough” to allow data received at the powering device connector **304***a* to be transmitted via the powered device connector **306** as described in further detail below.

(37) However, while described as providing a data “passthrough”, one of skill in the art in possession of the present disclosure will appreciate how the data communication subsystem that couples the powered device connector **306** to the powering device connector **304***a* may include processing elements for processing the data received at the powering device connector **304***a* and transmitted via the powered device connector **306** while remaining within the scope of the present disclosure as well. Furthermore, while the embodiments illustrated and discussed below only describe the data communication subsystem coupling the powered device connector **306** to the powering device connector **304***a*, other embodiments may include the data communication subsystem coupling the powered device connector **306** to any or all of the powering device connectors **304***b*-**304***c* as well while remaining within the scope of the present disclosure as well.

(38) In many of the embodiments discussed below, the powered device connector **306** is provided by an Ethernet connector, although one of skill in the art in possession of the present disclosure will appreciate how other types of powered device connectors may enable the teachings of the present disclosure and thus are envisioned as falling within its scope as well. Furthermore, while only a single powered device connector **306** is illustrated in FIG. **3** and discussed in many of the embodiments detailed below, one of skill in the art in possession of the present disclosure will appreciate how the chassis may include additional powered device connectors that operate similarly to the powered device connector **306** described below while remaining within the scope of the present disclosure as well.

(39) In the illustrated embodiment, the chassis **302** also houses a power transmission subsystem **308** that is coupled to each of the powering device connectors **304***a*-**304***c*, as well as to the powered device connector **306**. In an embodiment, the power transmission subsystem **308** may be provided by an integrated circuit device, a power combiner device, a Direct-Current-to-Direct-Current (DC-to-DC) converter device, and/or other devices that one of skill in the art in possession of the present disclosure will recognize may be configured to aggregate power received from any combination of the powering device connectors **304***a*-**304***c* as described below. Furthermore, in some embodiments

a parallel power circuit may be provided using transformers (e.g., to combine power from multiple powering device ports), diodes (e.g., to prevent current from flowing back to an power input), capacitors (e.g., to filter out noise), resistors (e.g., to load balance power and serve power to the powered device), and/or other power handling components that would be apparent to one of skill in the art in possession of the present disclosure. However, while described as being provided by particular devices, one of skill in the art in possession of the present disclosure will appreciate how the power transmission subsystem **308** may be provided using a variety of power transmission components that are configured to perform the power transmission functionality described below while remaining within the scope of the present disclosure as well.

(40) In some embodiments, the chassis **302** may house a processing system (not illustrated, but which may be similar to the processor **102** discussed above with reference to FIG. **1**) and a memory system (not illustrated, but which may be similar to the memory **114** discussed above with reference to FIG. **1**) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide an aggregated PoE controller engine **310** that is configured to perform the functionality of the aggregated PoE controller engines, aggregated PoE controller subsystems, and/or PoE power aggregation devices discussed below. However, while the aggregated PoE controller engine **310** is illustrated and described below as being included in the PoE power aggregation device **300** and configured to perform powering device configuration operations, power negotiation operations, and/or other PoE controller operations that would be apparent to one of skill in the art in possession of the present disclosure, other embodiments of the present disclosure may omit the aggregated PoE controller engine **310** from the PoE power aggregation device **300**, with the powering device configuration operations performed by a user, and the power negotiation operations and other PoE controller operations performed by the powering device and powered device via the PoE power aggregation device **300** (e.g., using the data “passthrough” provided by the PoE power aggregation device **300**), as described in further detail below.

(41) As illustrated, in embodiments in which the aggregated PoE controller engine **310** is included in the PoE power aggregation device **300**, the aggregated PoE controller engine **310** may be coupled to each of the powering device connectors **304***a*-**304***c* (e.g., via a coupling between each of the powering device connectors **304***a*-**304***c* and the processing system discussed above), the power transmission subsystem **308** (e.g., via a coupling between the powering transmission subsystem **308** and the processing system discussed above), and the powered device connector **306** (e.g., via a coupling between the powered device connector **306** and the processing system discussed above). Furthermore, the chassis **302** may also house a power subsystem **312** that is coupled to the power transmission subsystem **308**, and that may be coupled to any of the PoE power aggregation components in the PoE power aggregation device **300** and configured to receive power from the power transmission subsystem **308** and provide that power to those PoE power aggregation components in order to enable the PoE power aggregation functionality described above. However, while a specific PoE power aggregation device **300** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that PoE power aggregation devices (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the PoE power aggregation device **300**) may include a variety of components and/or component configurations for providing conventional device functionality, as well as the PoE power aggregation functionality discussed below, while remaining within the scope of the present disclosure as well.

(42) With reference to FIG. **4**A, a specific embodiment of a PoE power aggregation device **400** is illustrated that may provide the PoE power aggregation device **300** discussed above with reference to FIG. **3**. In the illustrated embodiment, the PoE power aggregation device **400** includes a chassis **402** that houses the components of the PoE power aggregation device **400**, only some of which are illustrated and described below. The chassis **402** includes a pair of male connectors **404***a* and **404***b*

that provide a specific example of the plurality of powering device connectors **304***a*-**304***c* discussed above with reference to FIG. **3** and that, as discussed below, may configure the PoE power aggregation device **400** to allow a conventional PoE powering device to power some PoE+ powered devices, or configure the PoE power aggregation device **400** to allow a PoE+ powering device to power some UPoE powered devices. In many of the embodiments discussed below, the powering device connectors **404***a* and **404***b* are provided by Ethernet connectors, although one of skill in the art in possession of the present disclosure will appreciate how other types of male connectors may enable the teachings of the present disclosure and thus are envisioned as falling within its scope as well.

(43) Furthermore, the chassis **402** also includes a female connector **406** that provides a specific example of the powered device connector discussed above with reference to FIG. **3**, with the female connector **406** coupled to the male connector **404***a* by cabling, traces in a circuit board that supports the female connector **406** and the male connector **404***a*, and/or other data communication subsystems that may operate as a data “passthrough” to allow data received at the male connector **404***a* to be transmitted via the female connector **406**. However, similarly as discussed above, while described as providing a data “passthrough”, one of skill in the art in possession of the present disclosure will appreciate how the data communication subsystem that couples the female connector **406** to the male connector **404***a* may include processing elements for processing the data received at the male connector **404***a* and transmitted via the female connector **406** while remaining within the scope of the present disclosure as well. Furthermore, while the embodiments illustrated and discussed below only describe the data communication subsystem coupling the female connector **406** to the male connector **404***a*, other embodiments may include the data communication subsystem coupling the female connector **406** to any or all of the male connectors **404***a* and **404***b* while remaining within the scope of the present disclosure as well. In many of the embodiments discussed below, the female connector **406** is provided by an Ethernet connector, although one of skill in the art in possession of the present disclosure will appreciate how other types of female connectors may enable the teachings of the present disclosure and thus are envisioned as falling within its scope as well.

(44) In the illustrated embodiment, the chassis **402** also houses a power transmission subsystem **408** that is coupled to each of the male connectors **404***a* and **404***b*, as well as to the female connector **406**. Similarly as discussed above, the power transmission subsystem **408** may be provided by an integrated circuit device, a power combiner device, a DC-to-DC converter device, and/or other devices that one of skill in the art in possession of the present disclosure will recognize may be configured to aggregate power received from any combination of the powering device connectors **304***a*-**304***c* as described below. As also described above, in some embodiments a parallel power circuit may be provided using transformers (e.g., to combine power from multiple powering device ports), diodes (e.g., to prevent current from flowing back to an power input), capacitors (e.g., to filter out noise), resistors (e.g., to load balance power and serve power to the powered device), and/or other power handling components known in the art. However, while described as being provided by particular devices, one of skill in the art in possession of the present disclosure will appreciate how the power transmission subsystem **408** may be provided using a variety of power transmission components that are configured to perform the power transmission functionality described below while remaining within the scope of the present disclosure as well.

(45) In some embodiments, the chassis **402** may house a processing system (not illustrated, but which may be similar to the processor **102** discussed above with reference to FIG. **1**) and a memory system (not illustrated, but which may be similar to the memory **114** discussed above with reference to FIG. **1**) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide an aggregated PoE controller engine **410** that is configured to perform the functionality of the aggregated PoE controller engines, aggregated PoE controller subsystems, and/or PoE power aggregation devices

discussed below. However, while the aggregated PoE controller engine **410** is illustrated and described below as being included in the PoE power aggregation device **400** and configured to perform powering device configuration operations, power negotiation operations, and/or other PoE controller operations that would be apparent to one of skill in the art in possession of the present disclosure, other embodiments of the present disclosure may omit the aggregated PoE controller engine **410** from the PoE power aggregation device **400**, with the powering device configuration operations performed by a user, and the power negotiation operations and other PoE controller operations performed by the powering device and powered device via the PoE power aggregation device **400** (e.g., using the data “passthrough” provided by the PoE power aggregation device **400**), as described in further detail below.

(46) As illustrated, in embodiments that include the aggregated PoE controller engine **410**, the aggregated PoE controller engine **410** may be coupled to each of the male connectors **404***a* and **404***b* (e.g., via a coupling between each of the male connectors **404***a* and **404***b* and the processing system discussed above), the power transmission subsystem **408** (e.g., via a coupling between the powering transmission subsystem **408** and the processing system discussed above), and the female connector **406** (e.g., via a coupling between the female connector **406** and the processing system discussed above). Furthermore, the chassis **402** may also house a power subsystem **412** that is coupled to the power transmission subsystem **408**, and that may be coupled to any of the PoE power aggregation components in the PoE power aggregation device **400** and configured to receive power from the power transmission subsystem **408** and provide that power to those PoE power aggregation components in order to enable the PoE power aggregation functionality described above.

(47) With reference to FIG. **4**B, a specific example of the PoE power aggregation device **400** is illustrated that provides the male connectors **404***a* and **404***b* on the chassis **402** in a “side-by-side” orientation **404** that one of skill in the art in possession of the present disclosure will appreciate may be configured to allow the male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that are provided in the same row (e.g., a “top” row of powering device ports or a “bottom” row of powering device ports on that powering device). With reference to FIG. **4**C, another specific example of the PoE power aggregation device **400** is illustrated that provides the male connectors **404***a* and **404***b* on the chassis **402** in a “stacked” orientation **406** with the male connector **404***b* inverted relative to the male connector **404***a*, which one of skill in the art in possession of the present disclosure will appreciate may be configured to allow the male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that are provided in different adjacent rows (e.g., a “top” row of powering device ports, and a “bottom” row of powering device ports on that powering device that are inverted relative to the “top” row of powering device ports).

(48) Furthermore, while the embodiment of FIG. **4**C includes the male connector **404***b* inverted relative to the male connector **404***a*, one of skill in the art in possession of the present disclosure will appreciate that the male connector **404***b* need not be inverted relative to the male connector **404***a* when the PoE power aggregation device **400** is configured for use with powering devices that do not invert their “bottom” row of powering device ports as described above. However, while specific examples of the PoE power aggregation device **400** have been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that PoE power aggregation devices (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the PoE power aggregation device **400**) may include a variety of components and/or component configurations for providing conventional device functionality, as well as the PoE power aggregation functionality discussed below, while remaining within the scope of the present disclosure as well.

(49) With reference to FIGS. **5**A, **5**B, **5**C, and **5**D, an embodiment of a PoE power aggregation device **500** is illustrated that provides a specific example of the PoE power aggregation device **400**

discussed above with reference to FIG. **4**A with configurable powering device connectors, and similar components in the PoE power aggregation devices **400** and **500** have been provided with the same reference numbers. The PoE power aggregation device **500** includes a first chassis portion **502***a* and a second chassis portion **502***b* that replace the chassis **402** discussed above with reference to FIG. **4**A, with the first chassis portion **502***a* including the male connector **404***a* and the female connector **410**, and the second chassis portion **502***b* including the male connector **404***b*. With reference to FIGS. **5**A and **5**B, the PoE power aggregation device **500** is illustrated in first powering device connector configuration that provides the male connectors **404***a* and **404***b* in a “side-by-side” orientation **504** that one of skill in the art in possession of the present disclosure will appreciate may be configured to allow the male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that are provided in the same row (e.g., a “top” row of powering device ports or a “bottom” row of powering device ports on that powering device).

(50) Furthermore, the first chassis portion **502***a* is movably coupled to the second chassis portion **502***b* by a moveable coupling **506** such as the hinge illustrated and discussed below, or other movable coupling that one of skill in the art in possession of the present disclosure would recognize as providing the powering device connector reconfiguration functionality described below. As will be appreciated by one of skill in the art in possession of the present disclosure, a ribbon cable or other data conduits may be provided between the first chassis portion **502***a* and the second chassis portion **502***b* to provide the couplings between the components of the PoE power aggregation device **500** that are included in the first chassis portion **502***a*, and the components of the PoE power aggregation device **500** that are included in the second chassis portion **502***b*, while also allowing the relative movement of the first chassis portion **502***a* and the second chassis portion **502***b* described below.

(51) For example, with reference to FIGS. **5**C and **5**D, the second chassis portion **502***b* may be rotated via the moveable coupling **506** in a direction A to reconfigure the male connectors **404***a* and **404***b* from the “side-by-side” orientation **504** illustrated and described above with reference to FIGS. **5**A and **5**B to a “stacked” orientation **508** with the male connector **404***b* inverted relative to the male connector **404***a*, which one of skill in the art in possession of the present disclosure will appreciate may be configured to allow the male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that are provided in different adjacent rows (e.g., a “top” row of powering device ports, and a “bottom” row of powering device ports on that powering device that are inverted relative to the “top” row of powering device ports).

(52) While not illustrated or described in detail, one of skill in the art in possession of the present disclosure will appreciate how the PoE power aggregation device **500** may include orientation securing features that are configured to engage and secure the first chassis portion **502***a* and the second chassis portion **502***b* in either of the “side-by-side” orientation **504** and the “stacked” orientation **508**. However, while a specific example of the PoE power aggregation device including configurable powering device connectors has been illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how configurable powering device connectors may be provided on the PoE power aggregation device of the present disclosure in a variety of manners that will fall within the scope of the present disclosure as well.

(53) With reference to FIGS. **6**A and **6**B, an embodiment of a PoE power aggregation device **600** is illustrated that provides another specific example of the PoE power aggregation device **400** discussed above with reference to FIG. **4**A with configurable powering device connectors, and similar components in the PoE power aggregation devices **400** and **600** have been provided with the same reference numbers. In the illustrated embodiment, the PoE power aggregation device **600** includes the male connector **404***b* connected to a moveable coupling **602** such as the slidable coupling illustrated and described below, or other movable coupling that one of skill in the art in possession of the present disclosure would recognize as providing the powering device connector

reconfiguration functionality described below. As will be appreciated by one of skill in the art in possession of the present disclosure, any of a variety of flexible cabling, adjustable cable couplings, or other data conduits may be provided between the male connector **404***b* and the data and/or power couplings described above in order to allow the movement of the male connector **404***b* relative to the male connector **404***a* as described below.

(54) For example, with reference to FIG. **6**B, the male connector **404***b* may be moved via the moveable coupling **602** and along a direction B to adjust a distance between the male connectors **404***a* and **404***b*, which one of skill in the art in possession of the present disclosure will appreciate may be configured to allow the male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that may be spaced-apart by a variety of different distances that fall within a distance capable of being achieved between the male connectors **404***a* and **404***b* based on the range of motion of the male connector **404***b* via the movable coupling **602**. While not illustrated or described in detail, one of skill in the art in possession of the present disclosure will appreciate how the PoE power aggregation device **600** may include connector securing features that are configured to secure the male connector **404***b* relative to the male connector **404***a* and prevent relative movement between the two (i.e., when a desired spacing is achieved).

(55) With reference to FIGS. **6**C and **6**D, a specific example of the PoE power aggregation device **600** is illustrated that provides the male connectors **404***a* and **404***b* on the chassis **402** in the “side-by-side” orientation **404** discussed above, with the moveable coupling **602** allowing the distance between male connectors **404***a* and **404***b* in the “side-by-side” orientation **404** to be adjusted in order to allow those male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that are provided in the same row but that may be spaced apart by different distances. With reference to FIGS. **6**E and **6**F, a specific example of the PoE power aggregation device **600** is illustrated that provides the male connectors **404***a* and **404***b* on the chassis **402** in the “stacked” orientation **406** discussed above, with the moveable coupling **602** allowing the distance between male connectors **404***a* and **404***b* in the “stacked” orientation **404** to be adjusted in order to allow those male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that are provided in different adjacent rows but that may be spaced apart by different distances.

(56) With reference to FIGS. **7**A and **7**B, an embodiment of a PoE power aggregation device **700** is illustrated that provides another specific example of the PoE power aggregation device **400** discussed above with reference to FIG. **4**A with configurable powering device connectors, and similar components in the PoE power aggregation devices **400** and **700** have been provided with the same reference numbers. In the illustrated embodiment, the PoE power aggregation device **700** includes the male connectors **404***a* and **404***b* connected to movable couplings **702***a* and **702***b*, respectively, such as the slidable couplings illustrated and described below, or other movable coupling that one of skill in the art in possession of the present disclosure would recognize as providing the powering device connector reconfiguration functionality described below. As will be appreciated by one of skill in the art in possession of the present disclosure, any of a variety of flexible cabling, adjustable cable couplings, or other data conduits may be provided between the male connectors **404***a* and **404***b* and the data and/or power couplings described above in order to allow the movement of the male connectors **404***a* and **404***b* relative to the chassis **402** as described below.

(57) For example, with reference to FIG. **7**B, either or both of the male connectors **404***a* and **404***b* may be moved via their respective moveable couplings **702***a* and **702***b* and along a direction C to adjust a distance between either of the male connectors **404***a* and **404***b* and the chassis **402**, which one of skill in the art in possession of the present disclosure will appreciate may be configured to allow the male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that may be provided at different depths on that powering device that fall

within a distance capable of being achieved between the male connectors **404***a* and **404***b* based on the range of motion of the male connectors **404***a* and **404***b* via their respective moveable couplings **702***a* and **702***b*. While not illustrated or described in detail, one of skill in the art in possession of the present disclosure will appreciate how the PoE power aggregation device **700** may include connector securing features that are configured to secure the male connectors **404***a* and **404***b* relative to the chassis **402** and prevent relative movement between the two (i.e., when a desired spacing is achieved). As will be appreciated by one of skill in the art in possession of the present disclosure, the configurable powering device connectors on the PoE power aggregation device **700** may be provided in either of the "side-by-side" or "stacked" orientations described above while remaining within the scope of the present disclosure.

(58) With reference to FIG. **8**, an embodiment of a PoE power aggregation device **500** is illustrated that provides another specific example of the PoE power aggregation device **400** discussed above with reference to FIG. **4**A with configurable powering device connectors, and similar components in the PoE power aggregation devices **400** and **700** have been provided with the same reference numbers. In the illustrated embodiment, the PoE power aggregation device **700** includes the male connectors **404***a* and **404***b* connected to cables **800***a* and **800***b*, respectively, that may be provided by Ethernet cables in some embodiments, but that one of skill in the art in possession of the present disclosure will appreciate may be provided by other cables that are configured to transmit the data and power as described below. As will be appreciated by one of skill in the art in possession of the present disclosure, the male connectors **404***a* and **404***b* may be moved via their respective cables **800***a* and **800***b* to allow the male connectors **404***a* and **404***b* to be connected to corresponding powering device ports on a powering device that are spaced apart within a distance capable of being achieved between the male connectors **404***a* and **404***b* based on the length of their respective cables **800***a* and **800***b*. As will be appreciated by one of skill in the art in possession of the present disclosure, the cables **800***a* and **800***b* on the PoE power aggregation device **700** may allow the male connectors **404***a* and **404***b* to be connected to powering device ports that are provided in either of the "side-by-side" or "stacked" orientations described above while remaining within the scope of the present disclosure.

(59) However, while several specific examples of PoE power aggregation devices with configurable powering device connector functionality have been illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how combinations of the configurable powering device connector functionality described above for the PoE power aggregation devices **500**, **600**, **700**, and **800** may be provided to further enhance the configurability of the powering device connectors on the PoE power aggregation device of the present disclosure.

(60) With reference to FIG. **9**, a specific embodiment of a PoE power aggregation device **900** is illustrated that may provide the PoE power aggregation device **300** discussed above with reference to FIG. **3**. In the illustrated embodiment, the PoE power aggregation device **900** includes a chassis **902** that houses the components of the PoE power aggregation device **900**, only some of which are illustrated and described below. The chassis **902** includes a pair of female connectors **904***a* and **904***b* that provide a specific example of the plurality of powering device connectors **304***a*-**304***c* discussed above with reference to FIG. **3** and that, as discussed below, may configure the PoE power aggregation device **900** to allow a conventional PoE powering device to power some PoE+ powered devices, or configure the PoE power aggregation device **400** to allow a PoE+ powering device to power some UPoE powered devices. In many of the embodiments discussed below, the powering device connectors **904***a* and **904***b* are provided by Ethernet connectors, although one of skill in the art in possession of the present disclosure will appreciate how other types of female connectors may enable the teachings of the present disclosure and thus are envisioned as falling within its scope as well.

(61) Furthermore, the chassis **902** also includes a female connector **406** that provides a specific example of the powered device connector discussed above with reference to FIG. **3**, with the

female connector **906** coupled to the female connector **904***a* by cabling, traces in a circuit board that supports the female connector **906** and the female connector **904***a*, and/or other data communication subsystems that may operate as a data "passthrough" to allow data received at the female connector **904***a* to be transmitted via the female connector **906**. However, similarly as discussed above, while described as providing a data "passthrough", one of skill in the art in possession of the present disclosure will appreciate how the data communication subsystem that couples the female connector **906** to the female connector **904***a* may include processing elements for processing the data received at the female connector **904***a* and transmitted via the female connector **906** while remaining within the scope of the present disclosure as well. Furthermore, while the embodiments illustrated and discussed below only describe the data communication subsystem as coupling the female connector **906** to the female connector **904***a*, other embodiments may provide the data communication subsystem coupling the female connector **906** to any or all of the female connectors **904***a* and **904***b* while remaining within the scope of the present disclosure as well. In many of the embodiments discussed below, the female connector **906** is provided by an Ethernet connector, although one of skill in the art in possession of the present disclosure will appreciate how other types of female connectors may enable the teachings of the present disclosure and thus are envisioned as falling within its scope as well.

(62) In the illustrated embodiment, the chassis **902** also houses a power transmission subsystem **908** that is coupled to each of the female connectors **904***a* and **904***b*, as well as to the female connector **906**. Similarly as discussed above, the power transmission subsystem **408** may be provided by an integrated circuit device, a power combiner device, a DC-to-DC converter device, and/or other devices that one of skill in the art in possession of the present disclosure will recognize may be configured to aggregate power received from any combination of the powering device connectors **304***a*-**304***c* as described below. As also described above, in some embodiments a parallel power circuit may be provided using transformers (e.g., to combine power from multiple powering device ports), diodes (e.g., to prevent current from flowing back to an power input), capacitors (e.g., to filter out noise), resistors (e.g., to load balance power and serve power to the powered device), and/or other power handling components known in the art. However, while described as being provided by particular devices, one of skill in the art in possession of the present disclosure will appreciate how the power transmission subsystem **908** may be provided using a variety of power transmission components that are configured to perform the power transmission functionality described below while remaining within the scope of the present disclosure as well.

(63) In some embodiments, the chassis **902** may house a processing system (not illustrated, but which may be similar to the processor **102** discussed above with reference to FIG. **1**) and a memory system (not illustrated, but which may be similar to the memory **114** discussed above with reference to FIG. **1**) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide an aggregated PoE controller engine **910** that is configured to perform the functionality of the aggregated PoE controller engines, aggregated PoE controller subsystems, and/or PoE power aggregation devices discussed below. However, while the aggregated PoE controller engine **910** is illustrated and described below as being included in the PoE power aggregation device **900** and configured to perform powering device configuration operations, power negotiation operations, and/or other PoE controller operations that would be apparent to one of skill in the art in possession of the present disclosure, other embodiments of the present disclosure may omit the aggregated PoE controller engine **910** from the PoE power aggregation device **900**, with the powering device configuration operations performed by a user, and the power negotiation operations and other PoE controller operations performed by the powering device and powered device via the PoE power aggregation device **900** (e.g., using the data "passthrough" provided by the PoE power aggregation device **900**) as described in further detail below.

(64) As illustrated, in embodiments that include the aggregated PoE controller engine **910**, the

aggregated PoE controller engine **910** may be coupled to each of the female connectors **904***a* and **904***b* (e.g., via a coupling between each of the female connectors **904***a* and **904***b* and the processing system discussed above), the power transmission subsystem **908** (e.g., via a coupling between the powering transmission subsystem **908** and the processing system discussed above), and the female connector **906** (e.g., via a coupling between the female connector **906** and the processing system discussed above). Furthermore, the chassis **902** may also house a power subsystem **912** that is coupled to the power transmission subsystem **908**, and that may be coupled to any of the PoE power aggregation components in the PoE power aggregation device **900** and configured to receive power from the power transmission subsystem **908** and provide that power to those PoE power aggregation components in order to enable the PoE power aggregation functionality described above. While not described in detail below, one of skill in the art in possession of the present disclosure will appreciate how the PoE power aggregation device **900** may operate similarly to the PoE powering device **400** discussed above but with the male connectors **404***a* and **404***b* on the PoE powering device **400** replaced by the female connectors **904***a* and **904***b* on the PoE powering device **900** that are configured to be coupled to powering device ports on a powering device via cabling (e.g., Ethernet cables).

(65) With reference to FIG. **10**, an embodiment of a rackable PoE power aggregation system **1000** is illustrated. In the illustrated embodiment, the rackable PoE power aggregation system **1000** includes a chassis **1002** that, in specific examples, may be dimensioned such that it is configured to be housed in a 1 Rack Unit (RU) computing device housing in a rack that is configured to house computing devices (although different sized chassis for the rackable PoE power aggregation system **1000** (e.g., 2 RU or more, chassis that occupy half of a 1 RU computing device housing, etc.) are envisioned as falling within the scope of the present disclosure as well). In the illustrated embodiment, a plurality of the PoE power aggregation devices **900** are included in the chassis **1002** of the rackable PoE power aggregation system **1000**. As will be appreciated by one of skill in the art in possession of the present disclosure, the plurality of PoE power aggregation devices **900** need not each include their respective chassis **902** when provided in the chassis **1002** of the rackable PoE power aggregation system **1000**, may share a common aggregated PoE controller engine when provided in the chassis **1002** of the rackable PoE power aggregation system **1000** (e.g., rather than each including a respective aggregated PoE controller engine **910** as illustrated), and/or may be subject to other modifications relative to the “stand-alone” PoE power aggregation device **900** discussed above with reference to FIG. **9**. As such, the respective pair of female connectors **904***a* and **904***b* (i.e., powering device connectors), along with the respective female connector **906** (i.e., a powered device connector) on each of the PoE power aggregation devices **900** may be accessible on a surface of the rackable PoE power aggregation system **1000**, and the rackable PoE power aggregation system **1000** may be positioned in a rack (e.g., a 1 RU housing defined by the rack).

(66) Referring now to FIG. **11**, an embodiment of a method **1100** for aggregating PoE power from a powering device for use by a powered device is illustrated. As discussed below, the systems and methods of the present disclosure provide for the aggregation of respective power received from multiple powering device ports on a powering device, and the transmission of that respective power along with data received from at least one of those powering device ports to a powered device. For example, the PoE power aggregation system of the present disclosure may include a PoE power aggregation device coupling a powering device to a powered device. The PoE power aggregation device includes first and second powering device connectors coupled to first and second powering device ports on the powering device, respectively, and a powered device connector coupled to a powered device port on the powered device. The PoE power aggregation device receives data from the powering device via the connected first powering device port/first powering device connector, and transmits the data to the powered device through the connected powered device connector/powered device port. The PoE power aggregation device also receives first and second power from the powering device via the connected first powering device port/first powering device

connector and the connected second powering device port/second powering device connector, respectively, and transmits the first and second power to the powered device through the connected powered device connector/powered device port along with the data. As such, powering devices may power powered devices that require a power level that exceeds the maximum power available from any one of their powering device ports by aggregating power from a plurality of those powering device ports and providing that aggregated power to that powered device.

(67) The method **1100** begins at block **1102** where a PoE power aggregation device couples to a powering device via a plurality of powering device connectors, and couples to a powered device via a powered device connector. With reference to FIG. **12**, in an embodiment of block **1102**, the PoE power aggregation device **400** may be coupled to the powering device **200***b* by connecting the male connectors **404***a* and **404***b* that provide the powering device connectors on the PoE power aggregation device **400** to the powering device ports **202***a* and **202***b* (e.g., female powering device ports), respectively, on the powering device **200***b*. As will be appreciated by one of skill in the art in possession of the present disclosure, the male connectors **404***a* and **404***b* on the PoE power aggregation device **400** may be moved relative to each other as described above with reference to the PoE power aggregation devices **500**, **600**, **700**, and/or **800** in order to provide for the connection of those male connectors **404***a* and **404***b* to the powering device ports **202***a* and **202***b* as illustrated in FIG. **12**.

(68) Furthermore, while not illustrated and described in detail, one of skill in the art in possession of the present disclosure will recognize how, at block **1102**, the female connectors **904***a* and **904***b* that provide the powering device connectors on the PoE power aggregation device **900** discussed above with reference to FIG. **9** may be coupled to the powering device ports **202***a* and **202***b* (e.g., female powering device ports), respectively, on the powering device **200***b* via respective cables (e.g., Ethernet cables). Further still, while the PoE power aggregation devices are described above as being connected to the powering device **202***b*, one of skill in the art in possession of the present disclosure will appreciate how PoE power aggregation devices may be connected to the powering devices **202***a* and/or **202***c* while remaining within the scope of the present disclosure as well.

(69) With continued reference to FIG. **12**, in an embodiment of block **1102**, the PoE power aggregation device **400** may also be coupled to the powered device **206** by coupling the female connector **406** that provides the powered device connector on the PoE power aggregation device **400** to the powered device port **206***a* (e.g., a female powered device port) on the powered device **206** via a cable (e.g., an Ethernet cable). Furthermore, while not illustrated and described in detail, one of skill in the art in possession of the present disclosure will recognize how, at block **1102**, the female connector **906** that provides the powered device connector on the PoE power aggregation device **900** discussed above with reference to FIG. **9** may be coupled to the powered device port **206***a* (e.g., a female powered device port) on the powered device **206** via a cable (e.g., an Ethernet cable). Further still, while not illustrated and described in detail, one of skill in the art in possession of the present disclosure will appreciate how any of the PoE power aggregation devices **900** in the rackable PoE power aggregation system **1000** may couple a powering device to a powered device in a manner similar to that described for the stand-alone PoE power aggregation device **900**.

(70) As will be appreciated by one of skill in the art in possession of the present disclosure, the use of the PoE power aggregation device **400** to couple the powering device ports **202***a* and **202***b* on the powering device **202** to the powered device port **206***a* on the powered device **206** provides an embodiment of a PoE power aggregation system **1200** according to the teachings of the present disclosure. Furthermore, while the specific example provided herein only illustrates and describes two powering device connectors on the PoE power aggregation device connected to corresponding powering device ports on the powering device **202**, as discussed above three (or more) powering device connectors on the PoE power aggregation device may be connected to corresponding powering device ports on a powering device in order to aggregate increasing amounts of power for provisioning to powered device with relatively high power requirements.

(71) The method **1100** then proceeds to block **1104** where power for the powered device is negotiated with the powering device. As discussed above, some embodiments of present disclosure may provide the aggregated PoE controller engine **410** in the PoE power aggregation device **400** that is configured to perform powering device configuration operations, power negotiation operations, and/or other PoE controller operations that would be apparent to one of skill in the art in possession of the present disclosure. However, other embodiments of the present disclosure may omit the aggregated PoE controller engine **410** from the PoE power aggregation device **400**, with the powering device configuration operations performed by a user, and the power negotiation operations and other PoE controller operations performed by the powering device and the powered device via the PoE power aggregation device **400** (e.g., using the data “passthrough” provided by the PoE power aggregation device **400**)
(72) With reference to FIG. **13**, an embodiment of the PoE power aggregation device **400** is illustrated that does not include the aggregated PoE controller engine **410** discussed above with reference to FIG. **4**A. In such embodiments, following the coupling of the powering device **202** to the powered device **206** as discussed above with reference to FIG. **12**, a network administrator or other user may access the powered device **202** (e.g., via a Command Line Interface (CLI) or a variety of other management access techniques that would be apparent to one of skill in the art in possession of the present disclosure), configure a “logical PoE port” (referred to as “PoE-1” in the example below) on the powering device **202**, and add the powering device ports **202***a* and **202***b* (referred to as “Eth 1/1/1” and “Eth 1/1/2” in the example below) using, for example, the commands provided below via an Operating System (OS): OS(conf) #interface poe-channel PoE-1 OS(conf-if-PoE-1) #interface range Eth 1/1/1/-1/1/2 OS(conf-if-PoE-1) #exit
(73) As discussed below and as will be appreciated by one of skill in the art in possession of the present disclosure, the commands above may configure the powering device port/“Eth 1/1/1” to transmit both data and power, while configuring the powering device port/“Eth 1/1/2” to transmit power. In some embodiments, at block **1104** the network administrator or other user may then configure the powering device ports **202***a* and **202***b* on the powering device **202** to transmit respective amounts of power, the sum of which satisfy the power requirements of the powered device **206** (as determined by the network administrator or other user based on their knowledge of the powered device **206**).
(74) However, with reference to FIG. **13**B, in other embodiments and following the configuration of the powering device ports **202***a* and **202***b* on the powering device **202** by the network administrator or other user as described above, the powering device **202** and the powered device **206** may perform power negotiation operations **1300** via the PoE power aggregation device **400** in order to negotiate the power that will be provided by the powering device **202** to the powered device **206**. For example, the power negotiation operations **1300** may begin with the powering device **202** providing power having an initial voltage via its powering device port **202***a*, the male connector **404***a*, the power transmission subsystem **408**, the female connector **406**, and to the powered device **206** via its powered device port **206***a* in order to power the powered device **206**. Furthermore, as illustrated in FIG. **13**B, a subset of the power received from the powering device **202** at the power transmission subsystem **408** may be provided to the power subsystem **412** in order to power the PoE power aggregation device **400** as well.
(75) The power negotiation operations **1300** may then include the powering device **202** providing power having an increased voltage (i.e., relative to the initial voltage discussed above) via its powering device port **202***a*, the male connector **404***a*, the power transmission subsystem **408**, the female connector **406**, and to the powered device **206** via its powered device port **206***a*, and one of skill in the art in possession of the present disclosure will appreciate how the provisioning of the power with the increased voltage may allow the powering device **202** to determine a class of the powered device **206** via the powered device **206** drawing power according its class (e.g., a class **4** powered device). The power negotiation operations **1300** may then include the powering device

202 providing a type field request via its powering device port **202***a*, the male connector **404***a*, through the data "passthrough" discussed above to the female connector **406**, and to the powered device **206** via its powered device port **206***a*, and one of skill in the art in possession of the present disclosure will appreciate how the provisioning of the type field request may cause the powered device **206** to respond to the powering device **202** with its type (e.g., type **2**) via its powered device port **206***a*, the female connector **406**, through the data "passthrough" discussed above to the male connector **404***a*, and via the powering device port **202***a* on the powering device **202**.

(76) The power negotiation operations **1300** may then include the powering device **202** advertising its available power (e.g., 30 watts based on a maximum 15 watts available from each of the powering devices ports **202***a* and **202***b* in a specific example) via its powering device port **202***a*, the male connector **404***a*, through the data "passthrough" discussed above to the female connector **406**, and to the powered device **206** via its powered device port **206***a*, and one of skill in the art in possession of the present disclosure will appreciate how the advertisement of the available power may allow the powered device **206** to respond with a requested power (e.g., 25 watts) via its powered device port **206***a*, the female connector **406**, through the data "passthrough" discussed above to the male connector **404***a*, and via the powering device port **202***a* on the powering device **202**, allowing the powering device **202** to subsequently transmit that requested power via the powering device ports **202***a* and **202***b*. However, while specific power negotiation operations have been described, one of skill in the art in possession of the present disclosure will appreciate how the power that will be provided by the powering device to the powered device may be negotiated in a variety of manners that will fall within the scope of the present disclosure as well.

(77) However, with reference to FIG. **14**A, in some embodiments the aggregated PoE controller engine **410** in the PoE power aggregation device **400** may perform aggregated PoE configuration and negotiation operations **1400** in order to configure the powering device **202** for the aggregated PoE functionality described below, as well as negotiate the power that will be provided by the powering device **202** to the powered device **206**. For example, the aggregated PoE configuration and negotiation operations **1400** may begin with the powering device **202** providing power via its powering device port **202***a*, the male connector **404***a*, and to the power transmission subsystem **408**, with a subset of the power received from the powering device **202** at the power transmission subsystem **408** provided to the power subsystem **412** in order to power the PoE power aggregation device **400**. The aggregated PoE controller engine **410** may then use PoE negotiation techniques (e.g., like those discussed above between the powering device **202** and the powered device **206**) in order to negotiate the provisioning of an amount of power from the powering device **202** that one of skill in the art in possession of the present disclosure will recognize is sufficient to complete the aggregated PoE configuration and negotiation operations **1400** discussed below, determine a maximum amount of power available to the PoE power aggregation device **400** from the powering device **202**, as well as any other power details that would be apparent to one of skill in the art in possession of the present disclosure.

(78) The power negotiation operations **1400** may then include the aggregated PoE controller engine **410** using the power transmission subsystem **408** to provide power having an initial voltage via the female connector **406** and to the powered device **206** via its powered device port **206***a* in order to power the powered device **206**. The power negotiation operations **1400** may then include the aggregated PoE controller engine **410** using the power transmission subsystem **408** to provide power having an increased voltage (i.e., relative to the initial voltage discussed above) via the female connector **406** and to the powered device **206** via its powered device port **206***a*, and one of skill in the art in possession of the present disclosure will appreciate how the provisioning of the power with the increased voltage may allow the aggregated PoE controller engine **410** to determine a class of the powered device **206** via the powered device **206** drawing power according to its class (e.g., a class **4** powered device). The power negotiation operations **1400** may then include the aggregated PoE controller engine **410** providing a type field request via the female connector **406**

and to the powered device **206** via its powered device port **206***a*, and one of skill in the art in possession of the present disclosure will appreciate how the provisioning of the type field request may cause the powered device **206** to respond to the aggregated PoE controller engine **410** with its type (e.g., type **2**) via its powered device port **206***a* and the female connector **406**.

(79) The power negotiation operations **1400** may then include the aggregated PoE controller engine **410** advertising its available power (e.g., 30 watts based on a maximum 15 watts available from each of the powering devices ports **202***a* and **202***b* in a specific example, or any amount reported by or determined via the power negotiations between the PoE power aggregation device **400** and the powering device **202** as described above) via the female connector **406** and to the powered device **206** via its powered device port **206***a*, and one of skill in the art in possession of the present disclosure will appreciate how the advertisement of the available power may allow the powered device **206** to respond with a requested power (e.g., 25 watts) via its powered device port **206***a* and the female connector **406**. The aggregated PoE controller engine **410** may then use PoE negotiation techniques (e.g., like those discussed above between the powering device **202** and the powered device **206**) in order to negotiate the provisioning of the requested power from the powering device **202** to cause the powering device **202** to subsequently transmit that requested power via the powering device ports **202***a* and **202***b*. With reference to FIG. **14**B, in some embodiments, the aggregated PoE controller engine **410** may then perform power transmission subsystem configuration operations **1402** that may include configuring the power transmission subsystem **408** to use the power received via the powering device connectors **404***a* and **404***b* to provide the requested power via the female connector **206**.

(80) The method **1100** then proceeds to block **1106** where the PoE power aggregation device receives data from the powering device via a first powering device connector. In an embodiment, at block **1106**, the powering device **202** may transmit data via its powering device port **202***a* (e.g., based on the configuration of the powering device **202** discussed above with reference to block **1104**), and that data may be received by the PoE power aggregation device **400** via its male connector **404***a*.

(81) The method **1100** then proceeds to block **1108** where the PoE power aggregation device receives respective power from the powering device via the plurality of powering device connectors. In an embodiment, at block **1108**, the powering device **202** may transmit first power (e.g., 15 watts) via its powering device port **202***a* (e.g., based on the configuration of the powering device **202** discussed above with reference to block **1104**) such that the first power transmitted via the powering device port **202***a* is transmitted along with at least some of the data transmitted via the powering device port **202***a* at block **1106**, and may transmit second power (e.g., 15 watts) via its powering device port **202***b* (e.g., based on the configuration of the powering device **202** discussed above with reference to block **1104**), with that first and second power received by the PoE power aggregation device **400** via its male connectors **404***a* and **404***b*, respectively.

(82) The method **1100** then proceeds to block **1110** where the PoE power aggregation device transmits the data received via the first powering device connector and the respective power received via the plurality of powering device connectors via the powered device connector to the powered device. With reference to FIG. **15**, in an embodiment of block **1110**, the PoE power aggregation device **400** may perform data transmission operations **1500** that include transmitting the data received from the powering device **202** via the male connector **404***a* via the "data passthrough" discussed above and to the female connector **406** such that the data is provided to the powered device port **206***a* (e.g., via an Ethernet cable) and to the powered device **206**.

(83) Furthermore, the PoE power aggregation device **400** may also perform power transmission operations **1502***a* that include transmitting the first power (e.g., 15 watts) received from the powering device **202** via the male connector **404***a* to the power transmission subsystem **408**, power transmission operations **1502***b* that include transmitting the second power (e.g., 15 watts) received from the powering device **202** via the male connector **404***b* to the power transmission subsystem

408, power aggregation operations that include the power transmission system **408** aggregating the first power and second power received via the male connectors **404***a* and **404***b* to provide aggregated power (e.g., 30 watts), power transmission operations **1504** that include the power transmission subsystem **408** providing a portion of the aggregated power to the power subsystem **412** for use in powering the PoE power aggregation device **400**, and power transmission operations **1506** that include transmitting the requested power (i.e., the 25 watts of power requested by the powered device **206** at block **1104** in the example above) via the female connector **406** such that the requested power is provided to the powered device port **206***a* (e.g., via an Ethernet cable) and to the powered device **206** for use by the powered device **206** to operate.

(84) Thus, systems and methods have been described that provide for the aggregation of respective power received from multiple powering device ports on a powering device, and the transmission of that respective power along with data received from at least one of those powering device ports to a powered device. For example, the PoE power aggregation system of the present disclosure may include a PoE power aggregation device coupling a powering device to a powered device. The PoE power aggregation device includes first and second powering device connectors coupled to first and second powering device ports on the powering device, respectively, and a powered device connector coupled to a powered device port on the powered device. The PoE power aggregation device receives data from the powering device via the connected first powering device port/first powering device connector, and transmits the data to the powered device through the connected powered device connector/powered device port. The PoE power aggregation device also receives first and second power from the powering device via the connected first powering device port/first powering device connector and the connected second powering device port/second powering device connector, respectively, and transmits the first and second power to the powered device through the connected powered device connector/powered device port along with the data. As such, powering devices may power powered devices that require a power level that exceeds the maximum power available from any one of their powering device ports by aggregating power from a plurality of those powering device ports and providing that aggregated power to that powered device.

(85) Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be employed without a corresponding use of other features. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the embodiments disclosed herein.

Claims

1. A Power over Ethernet (PoE) power aggregation system, comprising: a powering device including a first powering device port and a second powering device port; a powered device including a powered device port; and a Power over Ethernet (PoE) power aggregation device that includes: a PoE power aggregation device chassis; a first powering device connector that is included on a first PoE power aggregation device chassis portion of the PoE power aggregation device chassis and that is coupled to the first powering device port included on the powering device; a second powering device connector that is included on a second PoE power aggregation device chassis portion of the PoE power aggregation device chassis and that is coupled to the second powering device port included on the powering device; a powered device connector that is included on the first PoE power aggregation device chassis portion of the PoE power aggregation device chassis and that is coupled to the powered device port included on the powered device; a hinge that moveably couples the first PoE power aggregation device chassis portion of the PoE power aggregation device chassis to the second PoE power aggregation device chassis portion of the PoE power aggregation device chassis and that is configured to move the first powering device

connector and the second powering device connector between a side-by-side orientation and a stacked orientation; a data transmission subsystem that is included in at least one of the first PoE power aggregation device chassis portion and the second PoE power aggregation device chassis portion of the PoE power aggregation device chassis, and that is coupled to the first powering device connector and the first powered device connector, wherein the data transmission subsystem is configured to receive data from the powering device via the first powering device port and through the first powering device connector, and transmit the data to the powered device through the powered device connector and via the powered device port; an aggregated PoE controller subsystem that is included in at least one of the first PoE power aggregation device chassis portion and the second PoE power aggregation device chassis portion of the PoE power aggregation device chassis, and that is coupled to the first powering device connector, the second powering device connector, and the powered device connector, wherein the aggregated PoE controller subsystem is configured to: negotiate, with the powering device via at least one of the first powering device connector and the second powering device connector, the provisioning of first power by the powering device through the first powering device connector and the provisioning of second power by the powering device through the second powering device connector; and negotiate, with the powered device via the powered device connector, the provisioning of third power through the powered device connector; and a power transmission subsystem that is included in at least one of the first PoE power aggregation device chassis portion and the second PoE power aggregation device chassis portion of the PoE power aggregation device chassis, and that is coupled to the first powering device connector, the second powering device connector, and the first powered device connector, wherein the power transmission subsystem is configured to receive the first power from the powering device via the first powering device port and through the first powering device connector, receive the second power from the powering device via the second powering device port and through the second powering device connector, and transmit the third power that includes at least some of the first power and the second power to the powered device through the powered device connector and via the powered device port along with the data.

2. The system of claim 1, wherein the first powering device connector is a male Ethernet connector, the second powering device connector is a male Ethernet connector, and the powered device connector is a female Ethernet connector.

3. The system of claim 1, wherein the side-by-side orientation of the first powering device connector and the second powering device connector configures the first powering device connector and the second powering device connector to connect to the first powering device port and the second powering device port in the same row on the powering device, and wherein the stacked orientation of the first powering device connector and the second powering device connector configures the first powering device connector and the second powering device connector to connect to the first powering device port and the second powering device port in adjacent rows on the powering device.

4. The system of claim 3, wherein the stacked orientation of the first powering device connector and the second powering device connector inverts the second powering device connector relative to the first powering device connector.

5. The system of claim 3, wherein the second powering device connector is configured to move through a channel defined by the second PoE power aggregation device chassis portion of the PoE power aggregation device chassis to adjust a distance between the first powering device connector and the second powering device connector.

6. The system of claim 1, wherein the first powering device connector is a female Ethernet connector, the second powering device connector is a female Ethernet connector, and the powered device connector is a female Ethernet connector.

7. A Power over Ethernet (PoE) power aggregation device, comprising: a chassis; a first powering device connector that is included on a first portion of the chassis; a second powering device

connector that is included on a second portion of the chassis; a powered device connector that is included on the first portion of the chassis; a hinge that moveably couples the first portion of the chassis to the second portion of the chassis and that is configured to move the first powering device connector and the second powering device connector between a side-by-side orientation and a stacked orientation; a data transmission subsystem that is included in at least one of the first portion and the second portion of the chassis and coupled to the first powering device connector and the first powered device connector, wherein the data transmission subsystem is configured to: receive data from a powering device through the first powering device connector; and transmit the data to a powered device through the powered device connector; a processing system that is included in at least one of the first portion and the second portion of the chassis and that is coupled to the first powering device connector, the second powering device connector, and the powered device connector; a memory system that is included in the chassis, that is coupled to the processing system, and that includes instructions that, when executed by the processing system, cause the processing system to provide an aggregated PoE controller engine that is configured to: negotiate, with the powering device via at least one of the first powering device connector and the second powering device connector, the provisioning of first power by the powering device through the first powering device connector and the provisioning of second power by the powering device through the second powering device connector; and negotiate, with the powered device via the powered device connector, the provisioning of third power through the powered device connector; and a power transmission subsystem that is included in at least one of the first portion and the second portion of the chassis and coupled to the first powering device connector, the second powering device connector, and the first powered device connector, wherein the power transmission subsystem is configured to: receive the first power from the powering device through the first powering device connector; receive the second power from the powering device through the second powering device connector; and transmit the third power that includes at least some of the first power and the second power to the powered device through the powered device connector along with the data.
8. The PoE power aggregation device of claim 7, wherein the first powering device connector is a male Ethernet connector, the second powering device connector is a male Ethernet connector, and the powered device connector is a female Ethernet connector.
9. The PoE power aggregation device of claim 7, wherein the side-by-side orientation of the first powering device connector and the second powering device connector configures the first powering device connector and the second powering device connector to connect to respective port in the same row on the powering device, and wherein the stacked orientation of the first powering device connector and the second powering device connector configures the first powering device connector and the second powering device connector to connect to respective ports in adjacent rows on the powering device.
10. The PoE power aggregation device of claim 9, wherein the stacked orientation of the first powering device connector and the second powering device connector inverts the second powering device connector relative to the first powering device connector.
11. The PoE power aggregation device of claim 9, wherein the second powering device connector is configured to move through a channel defined by the second portion of the chassis to adjust a distance between the first powering device connector and the second powering device connector.
12. The PoE power aggregation device of claim 7, wherein the first powering device connector is a female Ethernet connector, the second powering device connector is a female Ethernet connector, and the powered device connector is a female Ethernet connector.
13. The PoE power aggregation device of claim 7, wherein the aggregated PoE controller engine is configured to: configure the power transmission subsystem to use the at least some of the first power received from the powering device via the first powering device connector and the second power received from the powering device via the second powering device connector to provide the

third power to the powered device via the powered device connector.

14. A method for aggregating Power over Ethernet (PoE) power from a powering device for use by a powered device, comprising: moving, by a hinge that couples a first Power over Ethernet (PoE) power aggregation device chassis portion of a PoE power aggregation device to a second PoE power aggregation device chassis portion of the PoE power aggregation device, a first powering device connector included on the first PoE power aggregation device chassis portion of the PoE power aggregation device and a second powering device connector included on the second PoE power aggregation device chassis portion of the PoE power aggregation device between a side-by-side orientation and a stacked orientation; receiving, by the PoE power aggregation device, data from a powering device through the first powering device connector; transmitting, by the PoE power aggregation device, the data to a powered device through a powered device connector included on the first PoE power aggregation device chassis of the PoE power aggregation device; negotiating, by the PoE power aggregation device with the powering device via at least one of the first powering device connector and a second powering device connector, the provisioning of first power by the powering device through the first powering device connector and the provisioning of second power by the powering device through the second powering device connector: negotiating, by the PoE power aggregation device with the powered device via the powered device connector, the provisioning of third power through the powered device connector; receiving, by the PoE power aggregation device, the first power from the powering device through the first powering device connector; receiving, by the PoE power aggregation device, the second power from the powering device through a second powering device connector included on the PoE power aggregation device; and transmitting, by the PoE power aggregation device, the third power that includes at least some of the first power and the second power to the powered device through the powered device connector along with the data.

15. The method of claim 14, wherein the first powering device connector is a male Ethernet connector, the second powering device connector is a male Ethernet connector, and the powered device connector is a female Ethernet connector.

16. The method of claim 14, wherein the side-by-side orientation of the first powering device connector and the second powering device connector configures the first powering device connector and the second powering device connector to connect to the first powering device port and the second powering device port in the same row on the powering device, and wherein the stacked orientation of the first powering device connector and the second powering device connector configures the first powering device connector and the second powering device connector to connect to the first powering device port and the second powering device port in adjacent rows on the powering device.

17. The method of claim 16, wherein the stacked orientation of the first powering device connector and the second powering device connector inverts the second powering device connector relative to the first powering device connector.

18. The method of claim 16, wherein the second powering device connector is moved through a channel defined by the second PoE power aggregation device chassis portion of the PoE power aggregation device to adjust a distance between the first powering device connector and the second powering device connector.

19. The method of claim 14, wherein the first powering device connector is a female Ethernet connector, the second powering device connector is a female Ethernet connector, and the powered device connector is a female Ethernet connector.

20. The method of claim 14, wherein the negotiating of the provisioning of third power through the powered device connector by the PoE power aggregation device with the powered device includes determining a class and a type of the powered device.